Sentiment Analysis of Financial News for Stock Price Prediction Using Pretrained and Fine-Tuned Transformer Models

Dustin M. Haggett
Department of Mechanical Engineering
Stevens Institute of Technology
Hoboken, NJ 07030
dhaggett@stevens.edu

Abstract—This paper tests whether financial news sentiment helps predict stock returns, or whether markets are too efficient for it to matter. We extract sentiment from approximately 20,000 financial news articles using two domain-specific transformer models—FinBERT and FLANG-RoBERTa—and explore two fine-tuning strategies: (1) a multimodal neural network combining frozen BERT embeddings with five lagged market features to predict daily log returns, and (2) a LoRA fine-tuned FinBERT classifier for binary price direction prediction. We also conduct formal econometric tests (OLS, Granger causality, VAR) on aggregated monthly sentiment against S&P 500 returns. The main finding: pretrained models extract sentiment effectively, but that signal does not translate into usable edge on its own. The LoRA fine-tuned FinBERT model achieves the best directional accuracy of 50.6% with an F1-score of 0.564, outperforming both pretrained baselines (∼49%) and the frozen multimodal model. However, out-of-sample R^2 for continuous return prediction remains negative, consistent with the norm for daily return prediction. Notably, FinBERT achieves 87.2% accuracy on financial sentiment classification [3], yet direction prediction accuracy remains near 50%—this 37-percentage-point gap quantifies the distance between NLP quality and market predictability. The results align with the semi-strong efficient market hypothesis.

Index Terms—financial NLP, sentiment analysis, transformer models, stock prediction, FinBERT, LoRA, multimodal learning

I. Introduction

Can financial news sentiment predict stock returns? The efficient market hypothesis (EMH) says no—public information gets priced in too fast [1]. But behavioral finance suggests sentiment-driven mispricing can create short-lived windows of predictability [2], and recent transformer-based models like FinBERT [3] and FLANG-RoBERTa [4] have made it feasible to extract nuanced sentiment from unstructured financial text at scale.

This paper tests that tension directly. We evaluate two pretrained financial language models for sentiment classification, fine-tune both a multimodal regression model and a LoRA-adapted FinBERT classifier for direction prediction, and run econometric analyses at the monthly level to assess the sentiment–market relationship. The core contribution is showing that sentiment alone is weak, but parameter-efficient fine-tuning with LoRA achieves the best directional accuracy—and being transparent about where the signal runs out.

We pursue four specific goals:

1) Evaluate pretrained financial LLMs (FinBERT, FLANG-RoBERTa) for sentiment classification and test whether their scores predict next-day price direction.
2) Fine-tune a multimodal neural network combining frozen BERT embeddings with five lagged market features (lagged returns, volatility, RSI, MACD) to predict continuous daily log returns.
3) Fine-tune FinBERT with LoRA for binary price direction classification, achieving parameter-efficient adaptation with only 0.53% of parameters trainable.
4) Analyze sentiment–market dynamics at the monthly level using OLS regression, Granger causality, and VAR to determine whether aggregated sentiment carries predictive information for broad market returns.

II. Data Description

A. Datasets

News corpus. 20,550 financial news articles published between January 2017 and December 2020, each containing a publication timestamp, headline, article body, and associated stock ticker. The corpus covers 381 unique tickers, with the most frequent being AMZN (1,333), GOOGL (1,294), META (961), AAPL (868), and BA (716). Fig. 1 shows the temporal distribution of article volume.

Price data. 603,840 daily closing price records for 480 S&P 500 constituents (including the SPX index) spanning January 2017 to December 2021.

B. Preprocessing

Log returns. Daily log returns were computed as $r_t = \log(P_t/P_{t-1})$, preferred over simple returns for their additivity across time and approximate normality [5].

Lagged and technical features. Five numerical features were derived per article: 1-day lagged return, 5-day lagged return, 20-day rolling volatility, 14-day RSI (Relative

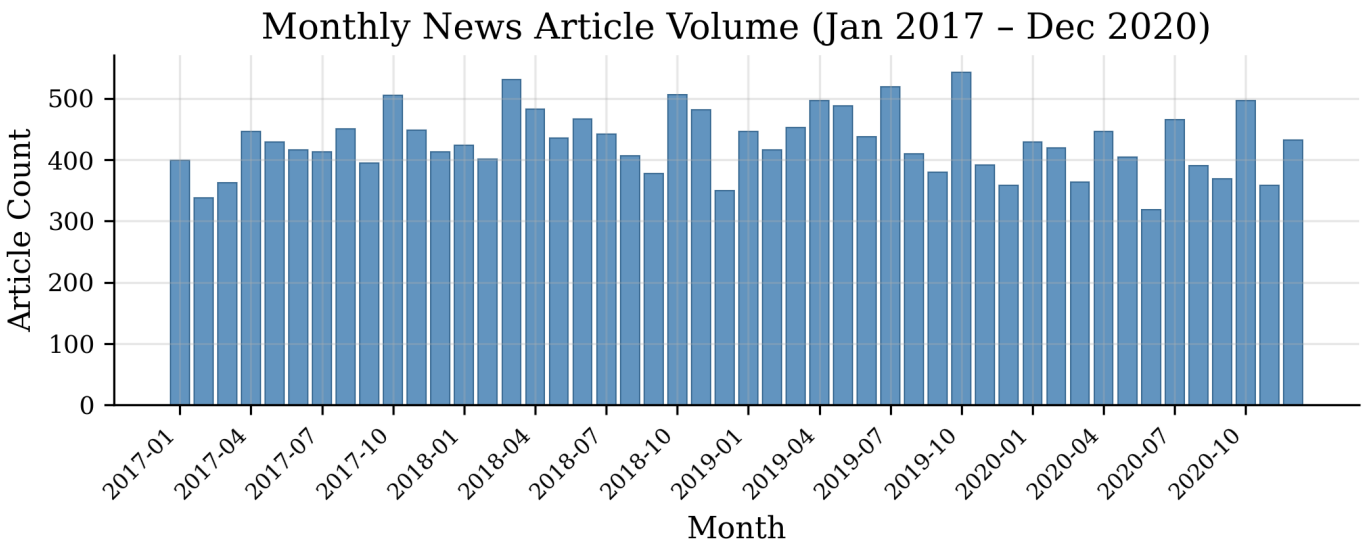

Fig. 1. Number of news articles per month across the full corpus (Jan 2017–Dec 2020). Coverage is relatively stable at 350–550 articles/month, with no sustained gaps.

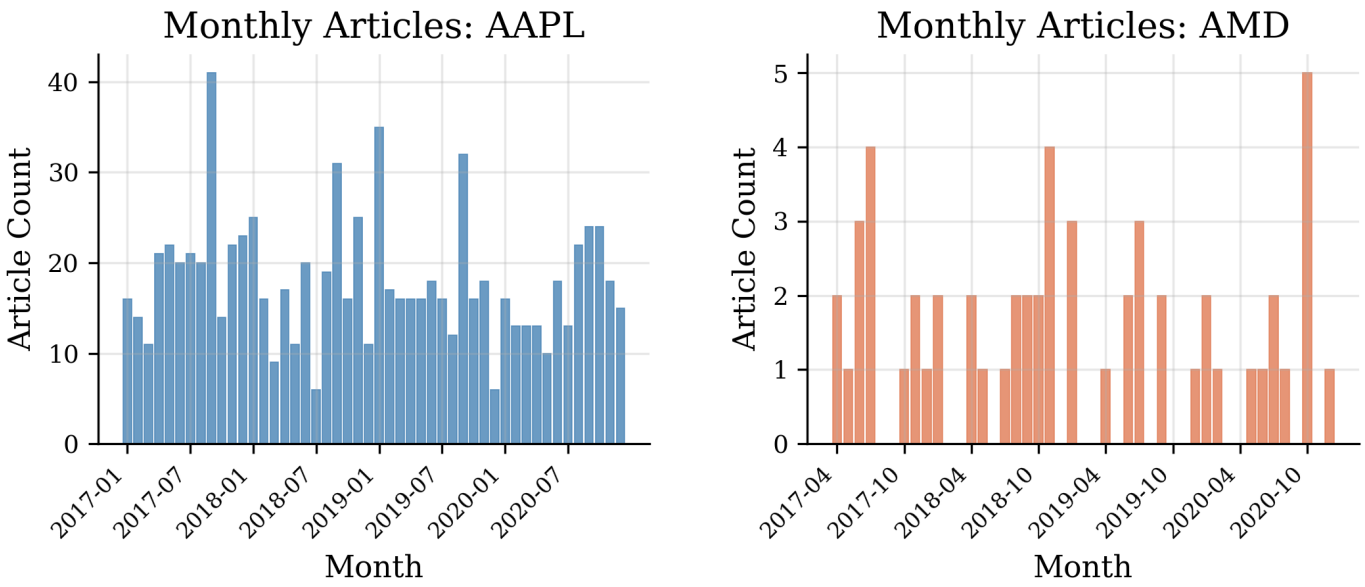

Fig. 2. Monthly article counts for two representative tickers: AAPL (high coverage, 15–40 articles/month) and AMD (sparse coverage, 1–5 articles/month). This heterogeneity motivates the use of article-level rather than ticker-aggregated modeling.

Strength Index), and MACD histogram (normalized by price). All features were lagged by one day to prevent look-ahead bias.

Z-score normalization. Both target returns and numerical features were standardized using training-set statistics only, preventing information leakage from the validation and test sets.

Net sentiment score. For pretrained models, we define a single net sentiment score:

$$s = P(\text{positive}) - P(\text{negative}) \tag{1}$$

ranging from -1 to $+1$, used for binary direction prediction ($s > 0 \Rightarrow$ up) and monthly aggregation.

Time-series split. A strict temporal split was applied: training on articles before January 1, 2019; validation on 2019; testing on 2020 onward—ensuring no look-ahead bias.

TABLE I
Pooling Strategy Comparison on the Test Set

Strategy	Val Loss	OOS R^2	Acc.	F1
[CLS] token	1.578	−0.014	0.479	0.328
Mean pool	1.564	−0.023	0.488	0.505
Mean+Max pool	1.576	−0.007	0.496	0.531

Mean pooling achieves the best validation loss with a balanced accuracy/F1 trade-off. The [CLS] F1 of 0.33 indicates near-complete collapse to single-class prediction.

III. Methodology

A. Pretrained Financial Language Models

FinBERT (yiyanghkust/finbert-tone) [3]: A BERT-base model fine-tuned on over 10,000 analyst reports for three-class financial tone classification, achieving approximately 87.2% accuracy on financial sentiment benchmarks. For each article, FinBERT produces class probabilities from which we derive the net sentiment score s per (1).

FLANG-RoBERTa (SALT-NLP/FLANG-Roberta) [4]: A RoBERTa model pretrained with preferential masking of financial domain terms, achieving 91.2% accuracy on the Financial PhraseBank benchmark. Since FLANG-RoBERTa is not natively a sentiment classifier, we extract mean-pooled hidden-state embeddings (768 dimensions) and train a lightweight logistic classifier on top.

B. Fine-Tuned Multimodal Model

We extract 768-dimensional mean-pooled embeddings from a frozen bert-base-uncased encoder for each article. These are concatenated with five numerical features (1-day lagged return, 5-day lagged return, 20-day volatility, RSI-14, and MACD histogram), producing a 773-dimensional input vector.

Why mean pooling over [CLS]? When the encoder is frozen, the [CLS] token is not optimized for the downstream task. Mean pooling aggregates information across all tokens, producing a more representative embedding [6]. Our ablation confirms this: mean pooling achieves the best validation loss and a substantially higher F1-score than [CLS] (0.50 vs. 0.33), as shown in TABLE I.

Architecture. The ReturnPredictor is a fully connected network with configurable depth (2–3 hidden layers), hidden sizes (128–512, halving at each layer), batch normalization, dropout ($p = 0.3$), and a configurable activation function. The output is a single scalar: the predicted standardized log return.

Training protocol. The model was trained with MSE loss and the Adam optimizer. Early stopping (patience = 10 epochs) and ReduceLROnPlateau scheduling prevented overfitting. A hyperparameter search over 7 configurations explored hidden size, depth, activation, learning rate, and weight decay. Fig. 3 shows the training and validation loss curves.

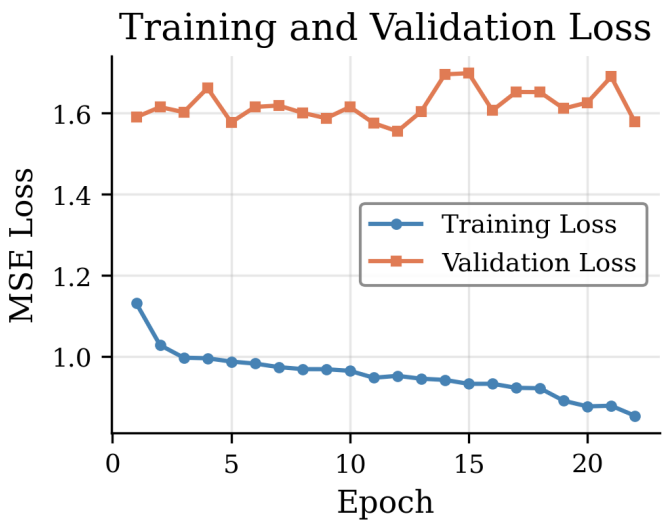

Fig. 3. Training and validation loss curves for the best multimodal model. Validation loss plateaus around epoch 3–4, a characteristic signature of the low signal-to-noise regime in daily financial prediction.

C. LoRA Fine-Tuned FinBERT

To bridge the gap between frozen pretrained models and full fine-tuning, we apply Low-Rank Adaptation (LoRA) to FinBERT for binary price direction classification (up/down). LoRA injects trainable low-rank decomposition matrices into the attention layers while keeping pretrained weights frozen, enabling parameter-efficient adaptation.

Configuration. We use LoRA rank 16 with $\alpha = 32$, adapting the query and value projection matrices in all 12 transformer layers. This yields 589,824 trainable parameters—only 0.53% of FinBERT's 110M total parameters. The model outputs two logits (down/up), trained with cross-entropy loss and class-balanced sample weights.

Training. Differential learning rates were used: 2×10^{-5} for LoRA parameters and 5×10^{-4} for the classification head. A warmup-plus-cosine schedule over 10 epochs provided stable optimization. Best validation accuracy of 51.7% was reached at epoch 2; validation performance degraded thereafter, confirming overfitting to noisy daily return labels.

D. Sentiment–Market Trend Analysis

Article-level sentiment was aggregated to monthly means and compared with S&P 500 next-month log returns using OLS regression, ADF stationarity tests, Granger causality (lags 1–3), VAR modeling, impulse response functions, and rolling-window OLS (12-month).

IV. Results

A. Binary Price Direction Prediction

All four models were evaluated on next-day up/down classification on the test set (2020 onward):

TABLE II
Model Comparison on Test Set (2020+)

Model	Acc.	F1	OOS R^2
Pretrained FinBERT	.498	.453	—
Pretrained FLANG-RoBERTa	.492	.457	—
Frozen BERT (multimodal)	.485	.529	−.014
LoRA FinBERT (classification)	.506	.564	—

The LoRA FinBERT model achieves the best accuracy and F1, confirming that parameter-efficient fine-tuning outperforms both frozen pretrained baselines and the multimodal regression approach for directional prediction.

FinBERT (pretrained). Accuracy of 49.8% with F1 of 0.453, barely below a coin flip.

FLANG-RoBERTa (pretrained). Accuracy of 49.2% with F1 of 0.457, comparable to FinBERT despite stronger domain pretraining. This indicates the bottleneck is signal strength, not extraction quality.

Frozen BERT (multimodal). Accuracy of 48.5% with the second-highest F1-score of 0.529. Integrating text embeddings with five lagged market features provides a meaningful F1 improvement over pretrained baselines.

LoRA FinBERT (classification). Achieved the highest accuracy of 50.6% and the highest F1-score of 0.564. Parameter-efficient fine-tuning allows the model to adapt FinBERT's representations to price direction prediction, yielding the best overall performance.

B. Continuous Return Prediction

The frozen BERT multimodal model produces an out-of-sample R^2 of -0.014 with MSE of 0.001400, meaning it underperforms a naive mean forecast. This is expected and well-documented [7]: daily individual-stock returns have an extremely low signal-to-noise ratio.

C. Sentiment vs. S&P 500 Returns

OLS regression. Monthly sentiment shows weak linear association with next-month S&P 500 returns. R^2 values are low across all sentiment measures. Fig. 4 and Fig. 5 show scatter plots and time-series overlays.

Granger causality. Results should be interpreted cautiously given the limited sample size (∼48–60 monthly observations).

VAR and impulse response. The VAR model captures bidirectional feedback between sentiment and returns. Impulse response functions show how a sentiment shock propagates over subsequent months.

Rolling-window OLS. The sentiment regression coefficient exhibits substantial temporal instability, particularly around the COVID-19 crash of early 2020.

D. Summary of Model Comparison

TABLE II summarizes the key metrics across all four approaches.

V. Discussion and Conclusion

The results paint a consistent picture: financial news contains real sentiment signal, but that signal is too weak for standalone return prediction without task-specific adaptation.

First, the gap between sentiment extraction quality and return prediction quality is striking. FinBERT achieves 87.2% accuracy on financial sentiment classification [3], yet direction prediction accuracy remains near 50%. This 37-percentage-point gap quantifies the distance between NLP quality and market predictability—the bottleneck is not sentiment extraction but market efficiency.

Second, parameter-efficient fine-tuning via LoRA proves to be the most effective approach. The LoRA FinBERT model achieves the best accuracy (50.6%) and F1-score (0.564) among all four models, despite adapting only 0.53% of the total parameters. This suggests that even modest task-specific adaptation of a domain-pretrained model can outperform both frozen pretrained baselines and multimodal fusion with a frozen encoder. However, the rapid overfitting after epoch 2 highlights the fundamental data limitation: daily return labels are inherently noisy.

Third, the frozen BERT multimodal approach provides complementary value. Adding five lagged market features to text embeddings produces the second-highest F1-score (0.529), confirming that text and market signals are partially complementary. However, the negative OOS R^2 (-0.014) reaffirms that daily stock returns remain effectively unpredictable.

Fourth, the monthly econometric analysis reveals regime dependence. The sentiment–return relationship shifts across market conditions, explaining why static models fail out-of-sample.

In summary: (1) pretrained LLMs extract sentiment effectively but direction accuracy hovers near 49%; (2) LoRA fine-tuning achieves the best performance at 50.6% accuracy and 0.564 F1 with only 0.53% of parameters trainable; (3) frozen BERT multimodal fusion achieves strong F1 (0.529) but 48.5% accuracy; (4) daily OOS R^2 is negative (-0.014), reflecting efficient price incorporation; (5) monthly sentiment tracks market direction but lacks statistical robustness; and (6) methodological rigor—time-series splitting, training-set-only normalization, pooling ablations, parameter-efficient fine-tuning—is essential to avoid overstated conclusions.

Practical implications. Sentiment signals are too weak for standalone trading but may add marginal value within a broader quantitative system—for longer holding periods, risk monitoring, or event detection. The LoRA approach demonstrates that parameter-efficient fine-tuning is a viable path for adapting large language models to financial prediction tasks.

Future work. Full end-to-end encoder fine-tuning, higher LoRA ranks, and weekly prediction horizons could improve performance. Portfolio-level evaluation—long-short

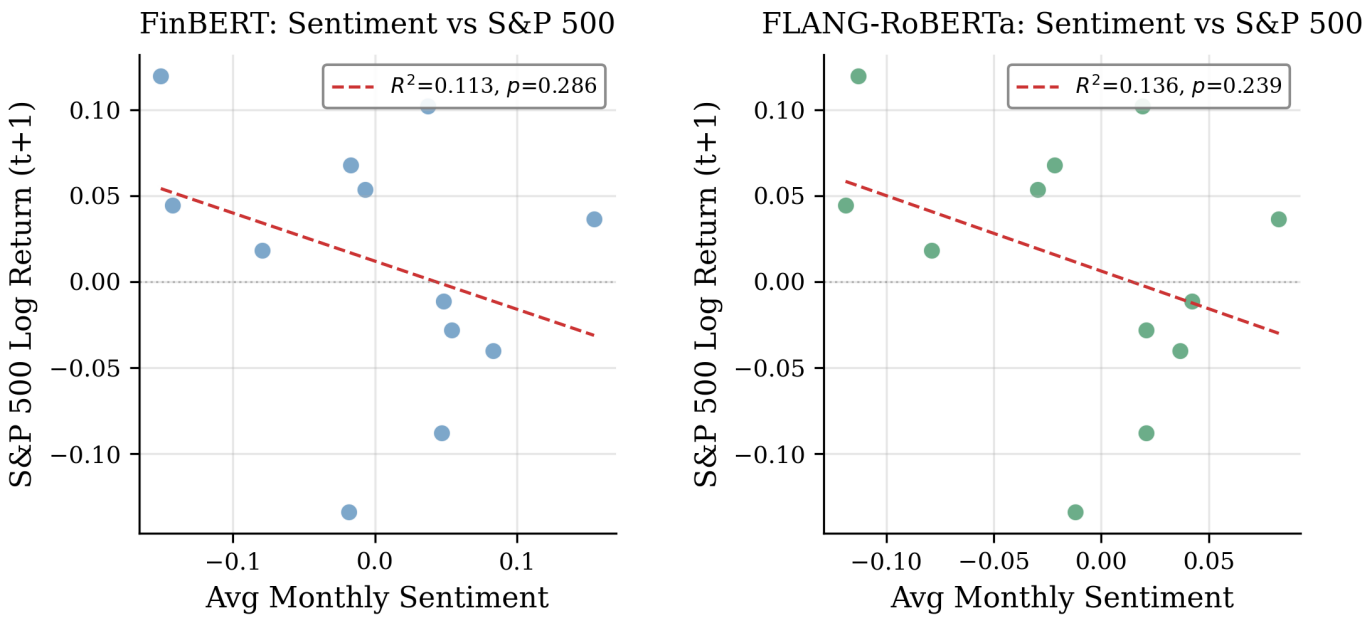

Fig. 4. Pretrained sentiment vs. S&P 500 returns. Left: FinBERT scatter plot of monthly average sentiment against next-month log return. Right: FLANG-RoBERTa scatter plot. Both models show weak linear association, with R^2 values below 0.15 and $p > 0.05$.

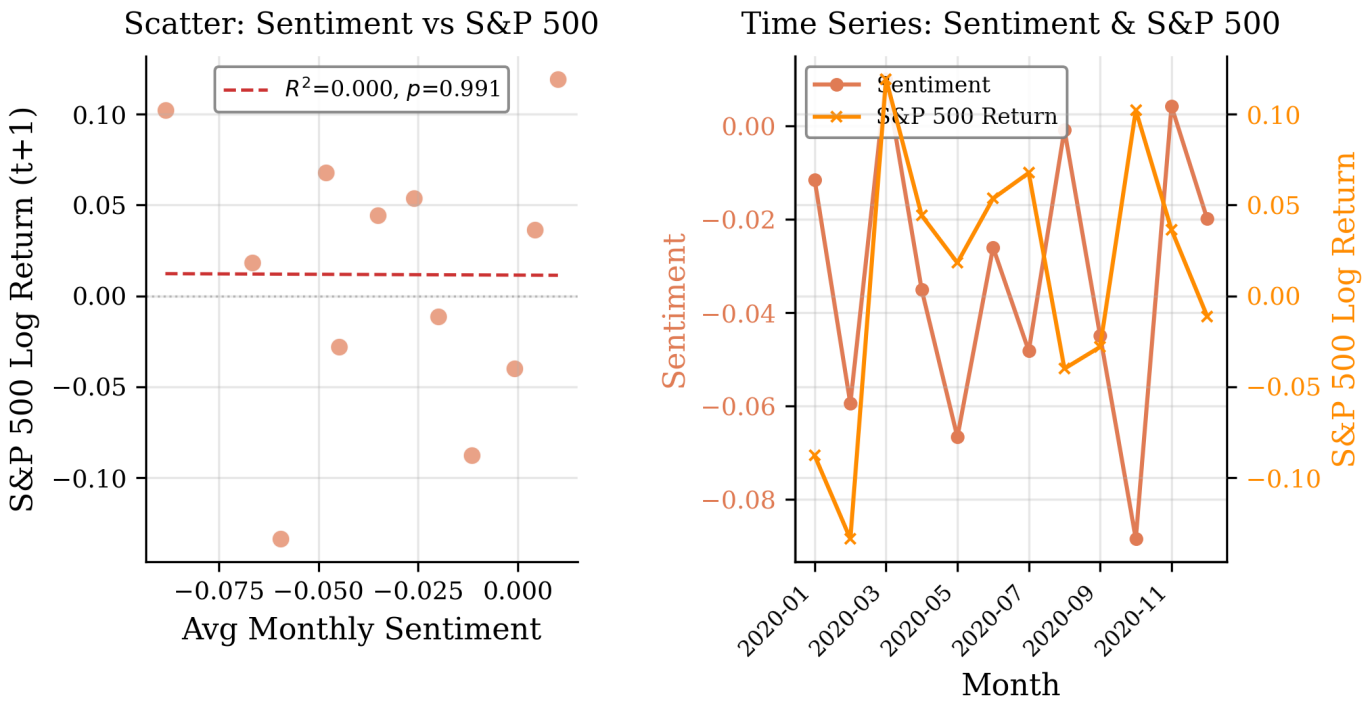

Fig. 5. Fine-tuned model sentiment vs. S&P 500 returns. Left: scatter plot with regression line. Right: time-series overlay showing directional alignment but weak magnitude correlation.

on sentiment deciles, measuring alpha after costs—would be the most practically relevant test.

Generative AI usage. Claude was used for code debugging, LaTeX formatting, and editorial feedback on draft revisions. Perplexity was used for background research.

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